

1 Simulation: confirmation in two balanced economies

We preserve the original stock-level baseline and add a smooth loading map that depends on all six characteristics. The historical experiment contains 3,500 environment/rank evaluations on 3,100 distinct economic replication seeds; the rank comparisons reuse baseline innovations. Its outputs and theoretical source bytes are archived unchanged. The following confirmation uses new seeds and a protocol frozen before performance was examined.

1.1 Economy and analytic reference

There are $N = 600$ stocks in triplets, six characteristics and three return factors. Group states start from their stationary Gaussian law and follow independent AR(1) dynamics with persistence 0.95. With $U_{g,d,t} = \Phi(S_{g,d,t})$, stock roles $r = 0, 1, 2$ have $Z_{g,r,d,t} = 2((U_{g,d,t} + r/3) \bmod 1) - 1$. These population ranks have continuous support; persistence is in characteristics, while factor shocks are iid. Define

$$\theta_\eta(z) = \pi(z_1 + 1) + \eta \sum_{d=2}^6 \sin\{\pi(z_d - z_1)\}, \quad \beta_\eta(z) = \sqrt{c_\beta} \begin{pmatrix} \sqrt{2/3} \cos \theta_\eta(z) \\ \sqrt{2/3} \sin \theta_\eta(z) \\ 1/\sqrt{3} \end{pmatrix}.$$

The original map uses $\eta = 0$ and the richer map uses $\eta = 0.35$. The common triplet phase rotates the first two loadings by $2\pi/3$ and preserves $\|\beta\|^2 = c_\beta$ and $B'_t B_t / N = (c_\beta/3)I_3$. Returns are $R_{t+1} = B_t F_{t+1} + \varepsilon_{t+1}$, with $F_{t+1} \sim N(\mu_F, I - \mu_F \mu_F')$ and independent $\varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2 I_N)$. We retain $c_\beta = 0.0016$, $\sigma_\varepsilon = 0.08$ and $\mu_F = (0.10, 0.06, 0.03)'$. For the paper's weights $W(Z_t)/N$, both maps have

$$W^*(z) = \frac{\beta_\eta(z)' \mu_F}{c_\beta/3 + \sigma_\varepsilon^2/N}, \quad q^* = \frac{c_\beta/3}{c_\beta/3 + \sigma_\varepsilon^2/N} \|\mu_F\|^2, \quad Q^* = 1 - q^*, \quad \text{SR}^* = \sqrt{q^*/(1 - q^*)}.$$

The original conditional-pricing and Gaussian-moment audits apply to both maps. An independent stock-dimensional solve checks E4, and analytic derivatives are compared with finite differences in all six coordinates. Smooth extensions with compactly supported cutoffs establish the required Sobolev membership; a finite Nyström norm is not used as a proof of infinite-RKHS membership. The operator sandwich underlying E5 is retained. Three economic factors do not imply rank three after integrating over characteristic states.

1.2 Frozen design, selectors and evaluation

The local-resource protocol uses 100 new paths per map on the practical grid

$$T \in \{60, 90, 120, 180, 240, 360, 540, 720, 1080, 1440\}.$$

The first 50 are coherently extended to $T \in \{2160, 3240, 4860, 7290\}$. Smoothness sensitivities $\nu \in \{0.5, 2.5\}$ use 50 paths per map on the practical grid. These counts are a prospective reduction from the proposed maximum 500/200/200 targets, recorded after a cost and projection pilot and before confirmation performance. Common random numbers pair maps, kernels and ranks; independent paths within an environment, rather than all environment evaluations, are the sampling units. The 400 environment evaluations cover 200 distinct map/innovation-index pairs: the two maps share 100 innovation seed indices, and alternative kernels reuse the corresponding economic paths. These are not 400 independent trajectories.

The response-one ridge estimator uses the actual Matérn kernel, unit diagonal, length scale one and rank 512 with independent nested Sobol anchors. Its uncentered second moment and penalty normalization are unchanged. Holdout25 uses the last quarter for validation and preserves its first-minimum tie rule. Rolling3 uses chronological train/validation endpoints at fractions (0.4, 0.6), (0.6, 0.8) and (0.8, 1), rounded down to integers. Validation loss is weighted by the number of observations; the selected policy is refitted on all T observations. Ties within 10^{-12} choose the larger penalty. Neither selector receives population moments, the optimum, oracle objects or test observations. A conservative one-standard-error variant is omitted because no dependent-data uncertainty rule was prospectively justified. The zero policy is an explicit benchmark with loss one, complexity zero and Sharpe convention zero; it is never silently added to a selector.

The independent stationary test and contiguous future $T + 1, \dots, T + 720$ each contain 720 observations. The latter uses the same long economic path, with no future data entering fitting. Sharpe is per period. Whole-path resampling preserves overlap across training lengths. Pointwise Monte Carlo intervals are conditional on the anchors and population quadrature, whose uncertainty is assessed separately.

For $b = 1 + 2\nu/6$, the separate theory experiment evaluates $\lambda_T(a) = as_{\text{ref}}T^{-b/(b+1)}$, $a \in \{0.25, 1, 4\}$, at the actual penalties, without grid snapping. The positive scale s_{ref} is fixed once per map/kernel from an independent population pilot. It uses population calibration information and is not an implementable tuning rule. The ensemble plugin loss oracle minimizes mean population regret on the fixed 96-point grid. Its Sharpe is evaluated at that loss-optimal penalty; it is not a Sharpe oracle. A five-fold oracle calibrates on four whole-path folds and evaluates on the fifth, exposing Monte Carlo selection optimism.

1.3 Evidence and numerical limits

For Baseline, mean population regret at $T = 1440$ is 0.006300 for holdout, 0.004419 for rolling validation, 0.003161 for the plugin loss oracle and 0.003161 for the cross-fitted loss oracle. These are observed comparisons, not a promise of a particular selector improvement.

For Rich6D, mean population regret at $T = 1440$ is 0.006559 for holdout, 0.004614 for rolling validation, 0.003389 for the plugin loss oracle and 0.003389 for the cross-fitted loss oracle. These are observed comparisons, not a promise of a particular selector improvement.

Mean, median and geometric-mean penalties are distinct because endpoint choices can dominate the mean. The tables and distribution figures report both endpoints separately. Negative sample OOS excess losses are retained on a signed linear scale. The exact population decomposition is

$$Q(\widehat{W}_\lambda) - Q^* = \{Q(W_\lambda) - Q^*\} + \|\widehat{W}_\lambda - W_\lambda\|_\Sigma^2 + 2\langle \widehat{W}_\lambda - W_\lambda, \Sigma W_\lambda - \mu \rangle.$$

The cross term is signed. Population bias includes the measured projection floor. Sample complexity is at most $\min(T, r_{\text{num}})$; population complexity divided by T need not be at most one.

Small exact temporal-Gram solves use $N \in \{30, 60\}$ and $T \in \{60, 120\}$, including the indispensable $T\lambda$ dual penalty. The practical-grid resolution audit compares ranks 256, 512, 1024 at fixed penalties on 50 paired paths, using common quadrature. Certification requires projection floor below 5 percent of relevant mean regret and absolute paired rank difference plus 1.96 MCSE below 5 percent. Fixed selected/theory policies are also evaluated with 8,192 and 32,768 groups at four independent quadrature seeds. Only regions passing the applicable rank and quadrature checks are described as resolved relative to the audited rank-1024 reference and quadrature. This finite comparison does not prove equality with the infinite kernel. False flags, near-interpolating regions and extra horizons without a corresponding rank check remain explicitly unresolved. No points or economic parameters are removed to secure a pass.

Spectrum calculations use 1,024,2,048,4,096 groups at three seeds, with independent dense-operator checks and partial eigenpair residuals. The windows 30–200,60–400,120–800 are fixed; unresolved indices are reported. A slope near $-b$ does not prove E5, and a distant finite-window fit does not disprove its analytic argument. The historical tolerance band is only descriptive.

The source condition is a theorem assumption. Neither smooth map is claimed to be least favourable, so neither must attain the minimax envelope with equality. The displayed theory paths, ex-post oracle fits and validation fits answer different questions. Their slopes and whole-path bootstrap intervals are reported without calibrating the economy or choosing windows after seeing the results. Cross-sectional size stays fixed within every rate experiment.

Table 1: Prospective confirmation at T=1440. Population risk uses independent quadrature; independent and contiguous-future test losses remain distinct. Sharpe is per period.

Map	Method	R	Loss	SR	Regret	OOS loss	Future loss
Baseline	Holdout	100	0.9921	0.09574	0.0063	0.9917	0.9923
Baseline	Rolling	100	0.9902	0.102	0.004419	0.9899	0.99
Baseline	Theory a=1	100	0.989	0.1075	0.003213	0.9883	0.9888
Baseline	Plugin oracle	100	0.9889	0.1074	0.003161	0.9882	0.9887
Baseline	Crossfit oracle	100	0.9889	0.1074	0.003161	0.9882	0.9887
Baseline	Zero	100	1	0	0.01422	1	1
Baseline	Affine	100	0.9985	0.05216	0.01272	0.9984	0.9988
Baseline	Equal weight	100	0.9992	0.02972	0.01337	0.999	0.9992
Rich6D	Holdout	100	0.9923	0.09405	0.006559	0.9922	0.9926
Rich6D	Rolling	100	0.9904	0.1012	0.004614	0.9902	0.9903
Rich6D	Theory a=1	100	0.9893	0.1064	0.003474	0.9887	0.9891
Rich6D	Plugin oracle	100	0.9892	0.1064	0.003389	0.9885	0.9889
Rich6D	Crossfit oracle	100	0.9892	0.1064	0.003389	0.9885	0.9889
Rich6D	Zero	100	1	0	0.01422	1	1
Rich6D	Affine	100	0.9986	0.05131	0.01277	0.9985	0.9987
Rich6D	Equal weight	100	0.9992	0.02972	0.01337	0.999	0.9992

Table 2: Separate penalty summaries and endpoint selection frequencies at T=1440. Frequencies are fractions; the zero policy is not eligible for validation.

Map	Selector	Mean lambda	Median	Geometric mean	Lower freq.	Upper freq.
Baseline	Holdout	0.0007122	1.274e-06	2.656e-06	0	0.07
Baseline	Rolling	0.0004034	1.624e-06	2.586e-06	0	0.04
Rich6D	Holdout	0.0009136	1e-06	2.708e-06	0	0.09
Rich6D	Rolling	0.0004032	1.624e-06	2.416e-06	0	0.04

Table 3: Exact theory penalties at T=7290 in the extended cohort. The scaled risk is descriptive; no equality with an asymptotic envelope is required. These extra horizons are not certified by the practical-grid rank audit.

Map	a	R	lambda	Regret	Scaled regret	C / T
Baseline	0.25	50	1.202e-07	0.001817	0.3776	0.004907
Baseline	1	50	4.808e-07	0.0009122	0.1896	0.00201
Baseline	4	50	1.923e-06	0.002023	0.4204	0.0008319
Rich6D	0.25	50	1.215e-07	0.001876	0.3898	0.004871
Rich6D	1	50	4.858e-07	0.001017	0.2113	0.001992
Rich6D	4	50	1.943e-06	0.002327	0.4835	0.0008217

Table 4: Numerical resolution at $T=1440$. The rank bound is absolute paired mean difference plus 1.96 MCSE. Both ratios must be at most 0.05, and the independent quadrature check must also pass. False flags retain unresolved results.

Map	Method	Floor / risk	Rank bound / risk	Rank pass	Quadrature pass
Baseline	Holdout	0.001589	0.01825	True	True
Baseline	Rolling	0.002294	0.01149	True	True
Baseline	Theory $a=1$	0.00332	0.01195	True	True
Rich6D	Holdout	0.008085	0.02133	True	True
Rich6D	Rolling	0.01196	0.01091	True	True
Rich6D	Theory $a=1$	0.01648	0.01209	True	True

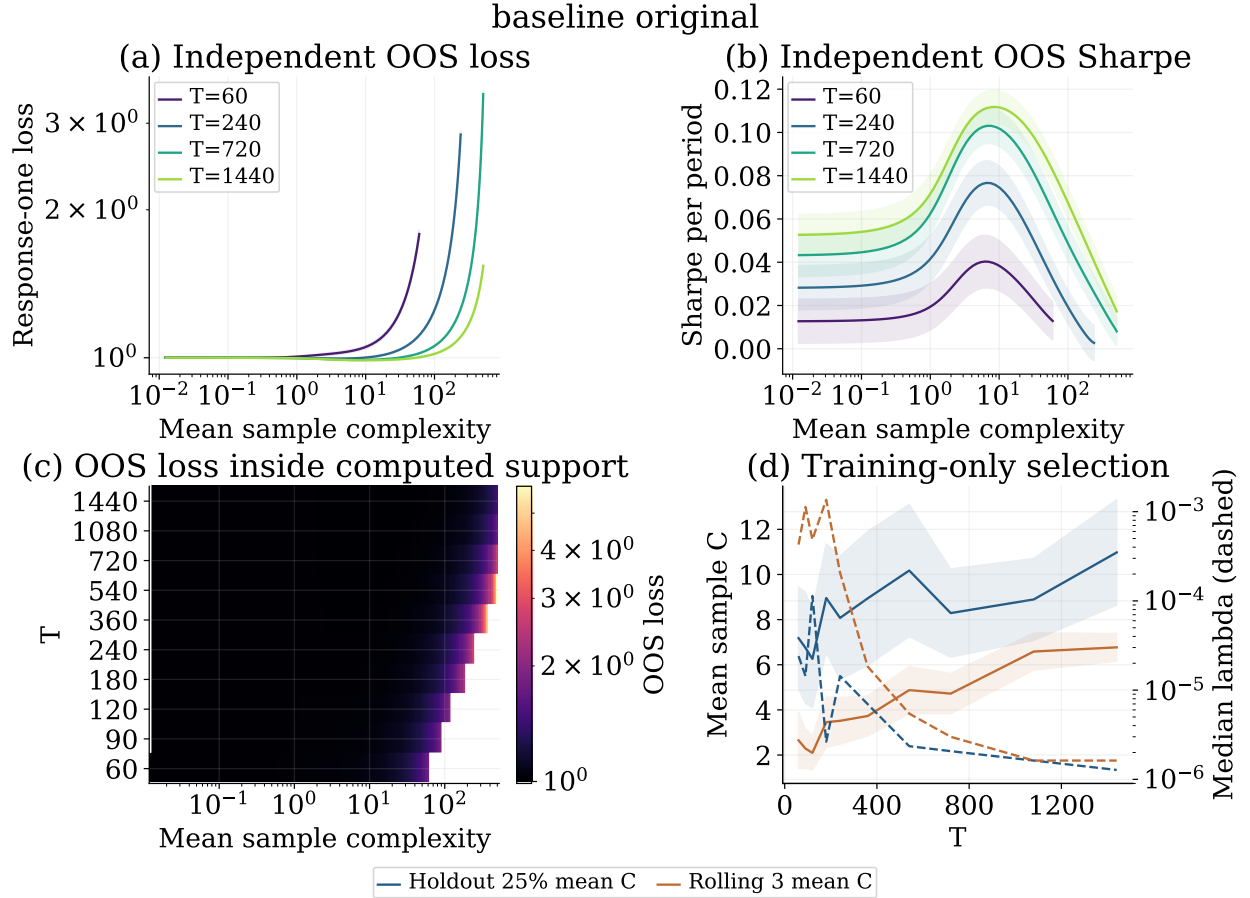


Figure 1: Baseline: full practical-grid OOS curves and training-only selection in the rank-512 representation. The entire curve is not certified as a full-kernel calculation. Bands are pointwise Monte Carlo intervals. Heatmaps interpolate only inside computed complexity support.

rich6d

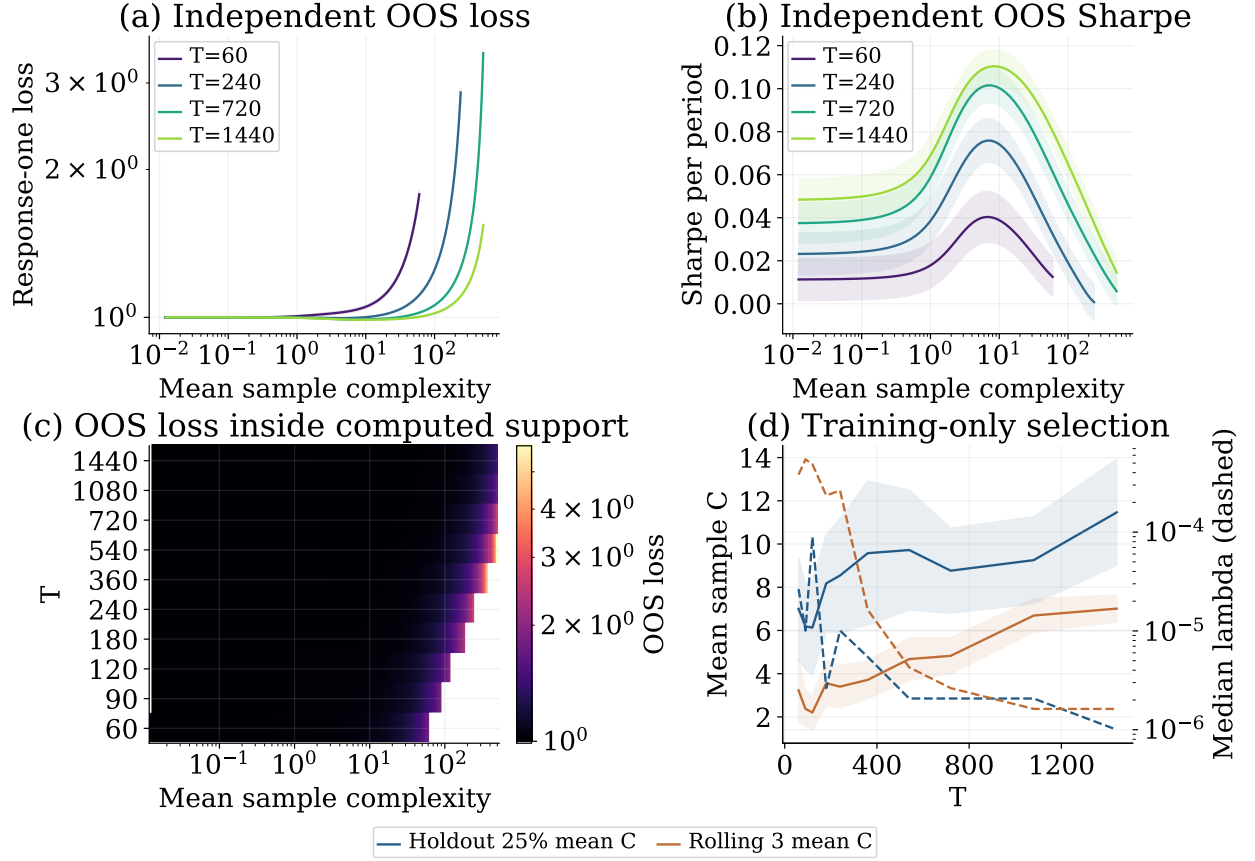


Figure 2: Rich6D: full practical-grid OOS curves and training-only selection in the rank-512 representation. The entire curve is not certified as a full-kernel calculation. Bands are pointwise Monte Carlo intervals. Heatmaps interpolate only inside computed complexity support.

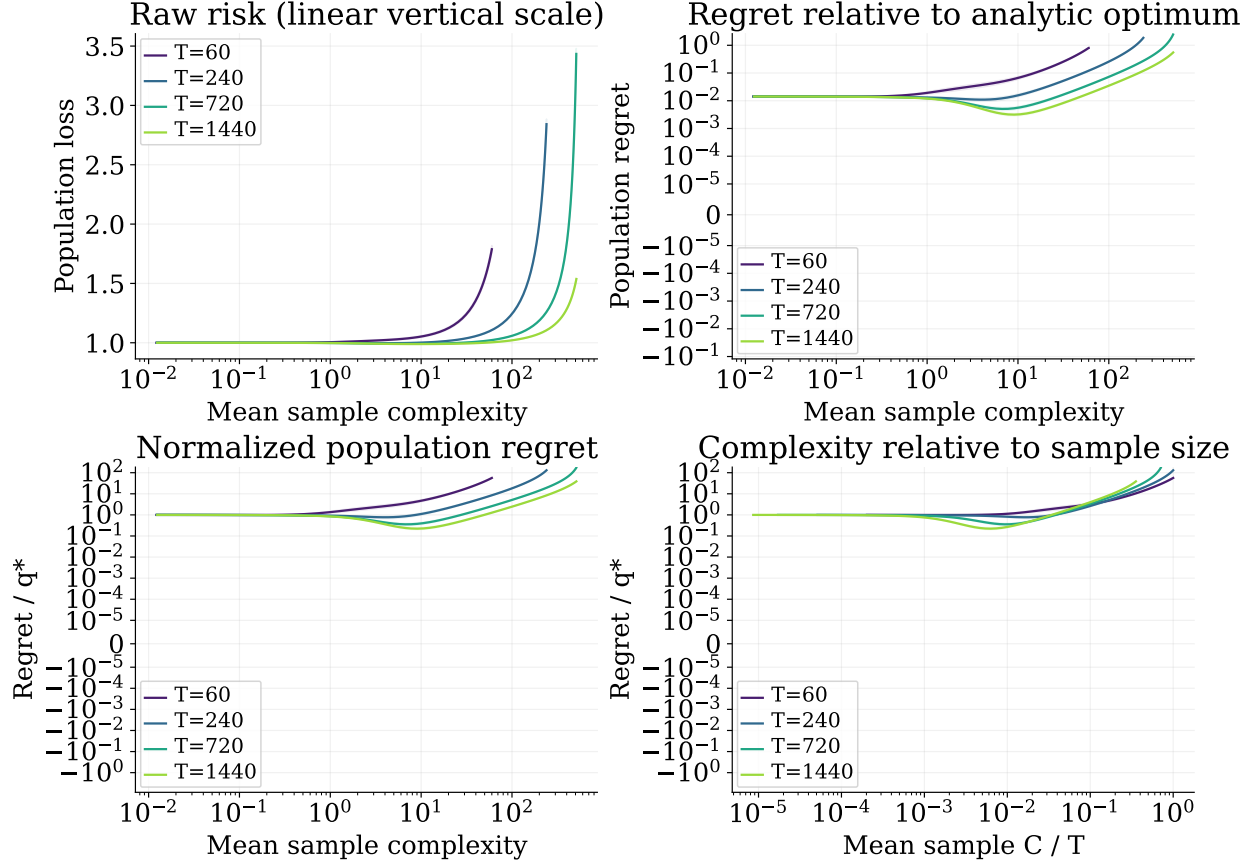


Figure 3: Baseline: raw population loss, regret and normalized regret against sample complexity and complexity per observation, at numerical rank 512. High-complexity regions failing the rank check remain diagnostic.

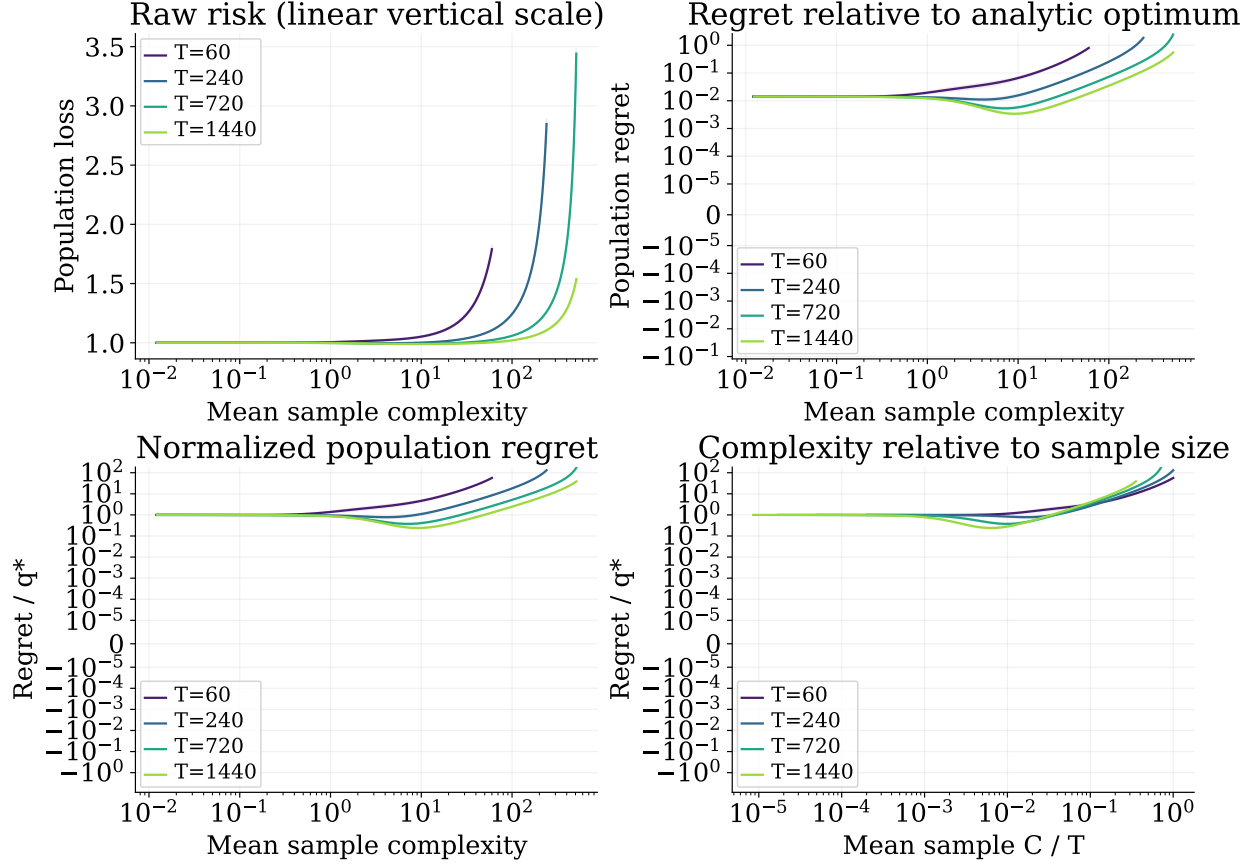


Figure 4: Rich6D: raw population loss, regret and normalized regret against sample complexity and complexity per observation, at numerical rank 512. High-complexity regions failing the rank check remain diagnostic.

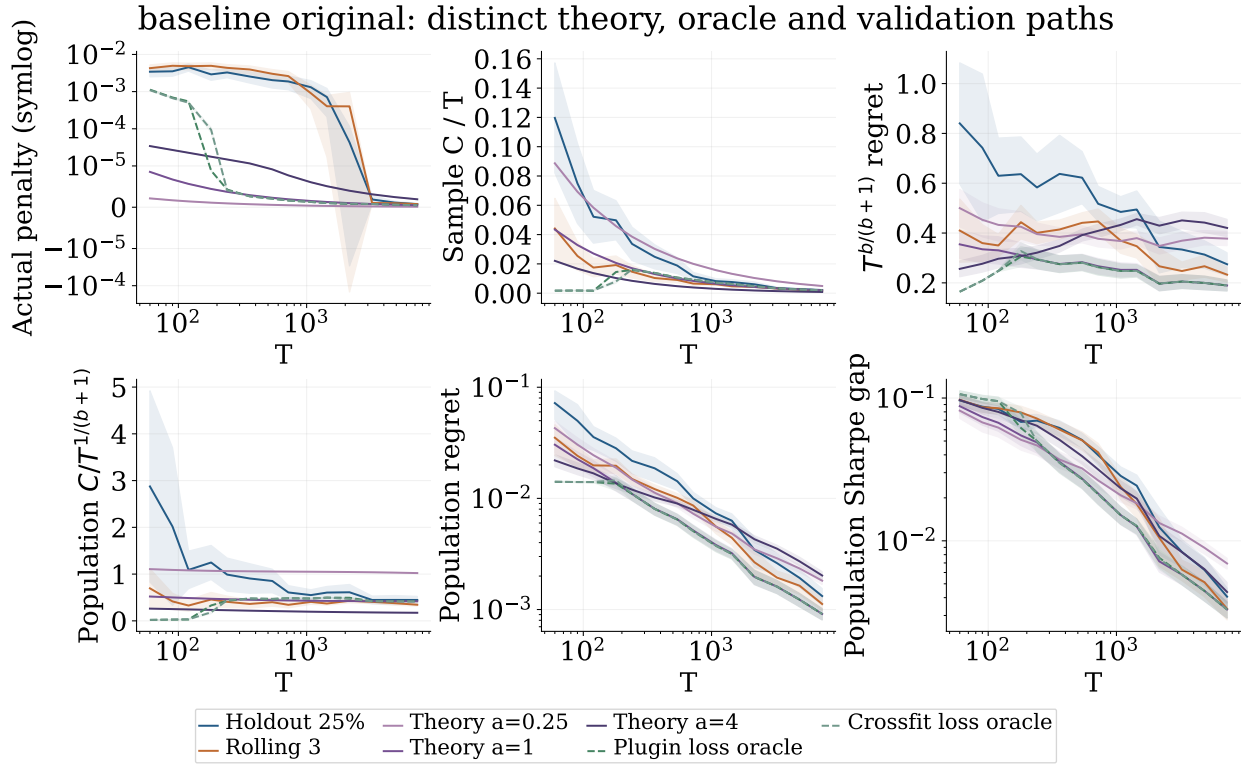


Figure 5: Baseline: distinct theory, oracle and validation paths. Larger training lengths use the 50-path extended cohort and retain finite-resolution qualifications. Panels labeled symlog preserve confidence intervals crossing zero, with linear threshold 10^{-5} . A negative penalty interval bound comes from the Gaussian Monte Carlo interval; selected penalties remain positive.

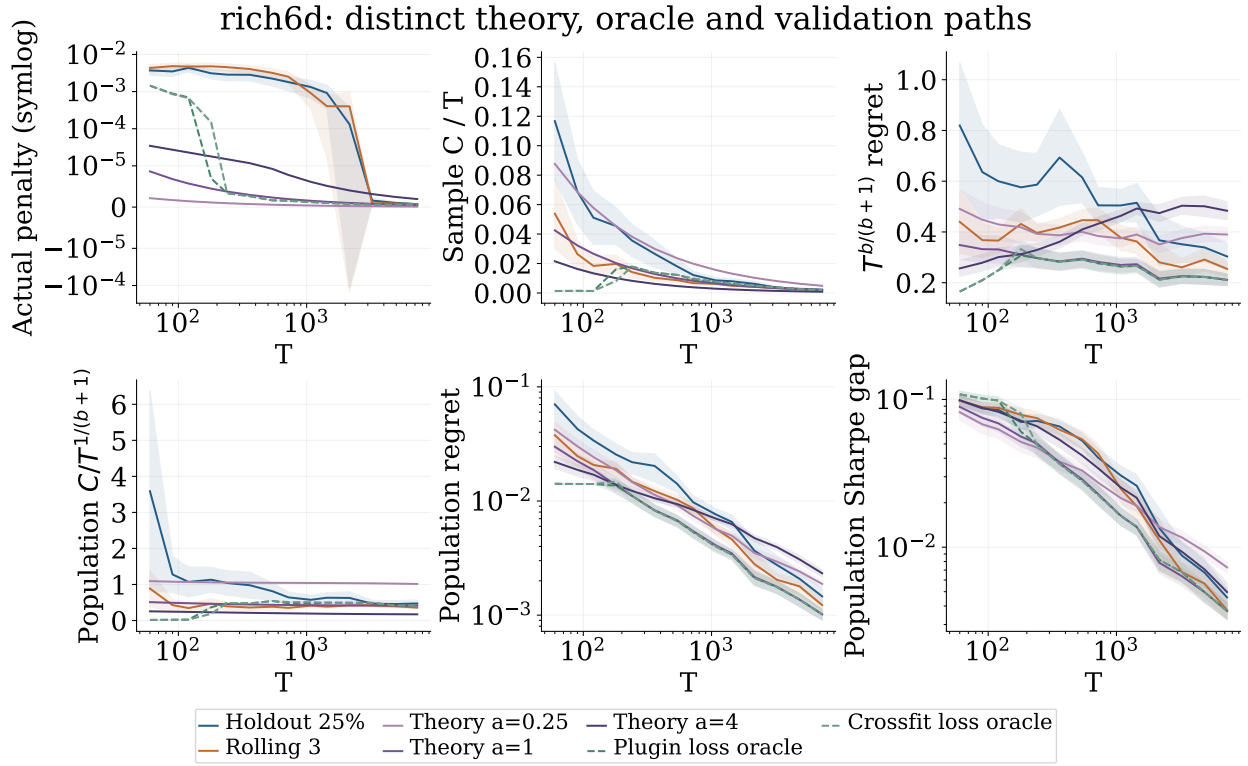


Figure 6: Rich6D: distinct theory, oracle and validation paths. Larger training lengths use the 50-path extended cohort and retain finite-resolution qualifications. Panels labeled symlog preserve confidence intervals crossing zero, with linear threshold 10^{-5} . A negative penalty interval bound comes from the Gaussian Monte Carlo interval; selected penalties remain positive.

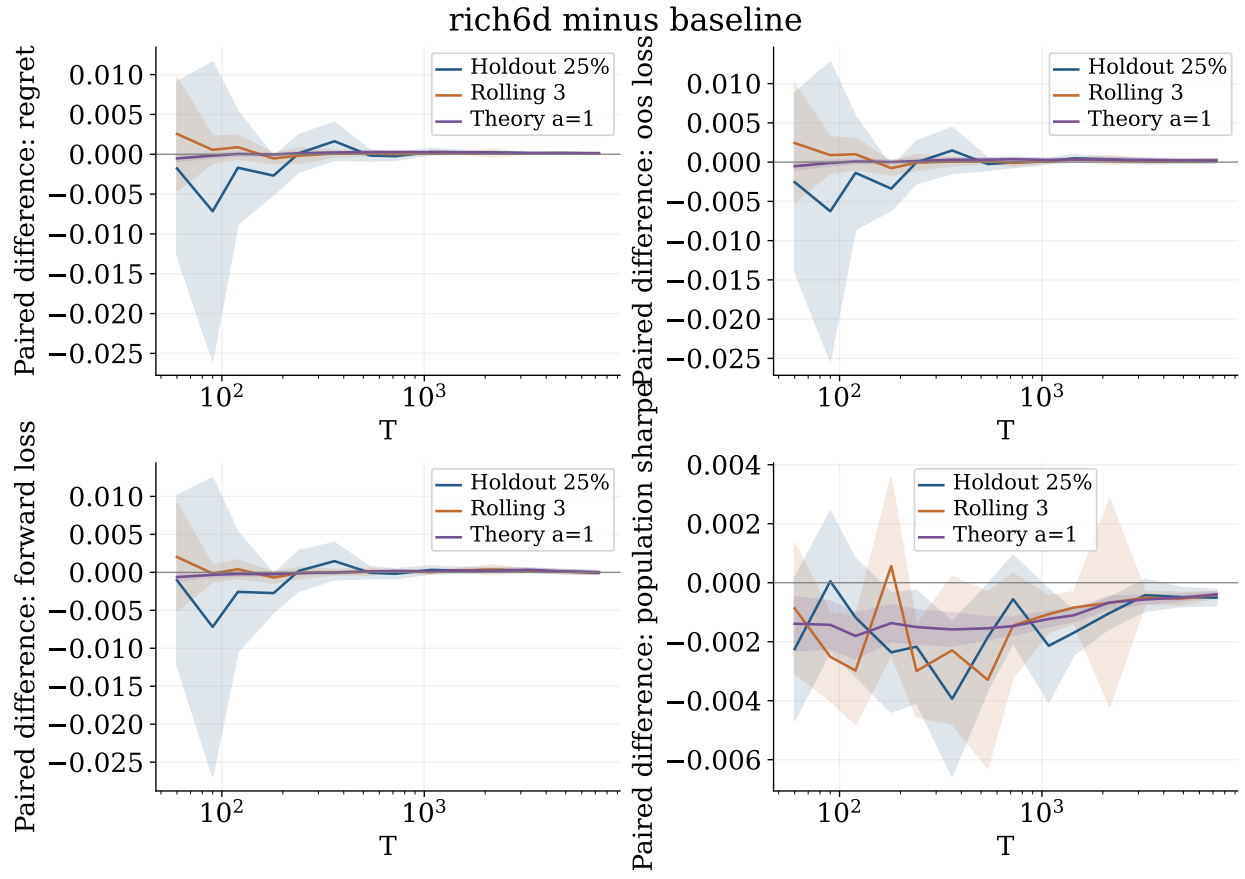


Figure 7: Paired rich-map minus original-map differences on common economic innovations; point-wise intervals use paired path differences as the sampling units.

Simulation confirmation appendix

All aggregates are reconstructed independently from saved path surfaces. The five stored training-only and theory policies also retain coefficients and actual test payoffs for population and OOS evaluation checks. Public synthetic archives contain checksums for each checkpoint; no licensed empirical observations are included. Historical checks reconstruct the original 3,500 evaluations and bootstrap intervals. That archive lacks stock returns and coefficients, so its aggregation check is not misrepresented as a fitting replay.

The historical empirical-rank robustness has finite support at fixed N and does not meet the baseline infinite-spectrum E5 interpretation. Its original one-coordinate loading map is preserved. Bounded heteroskedasticity does not impose baseline conditional pricing, and its historical regret reference is the finite-subspace optimum, not the unrestricted analytic baseline optimum. The licensed empirical exercise elsewhere in the paper is unchanged and separate from these artificial economies.

Table 5: Squared coordinate derivatives of the rich-map optimal policy. The analytic expectation covers the full uniform cube; Monte Carlo uses 131,072 independent points.

Coordinate	Analytic mean square	Uniform-cube MC mean	MCSE
1	316	317	1.148
2	14.82	14.81	0.04571
3	14.82	14.78	0.0457
4	14.82	14.85	0.0458
5	14.82	14.82	0.04578
6	14.82	14.85	0.04577

Table 6: Descriptive log-regret slopes for the exact theory path with $a=1$, in every fixed window and both declared cohorts. The extended cohort uses only its 50 complete paths at all training lengths. Bootstrap intervals resample whole paths; the theoretical exponent is an upper-envelope benchmark, not a required equality. All other path and metric fits are retained in the machine-readable [rate tables](#).

Map	Cohort	Window	R	Slope	OLS SE	Boot. low	Boot. high
Baseline	practical	full	100	-0.7102	0.007075	-0.7684	-0.6466
Baseline	practical	upper half	100	-0.6839	0.02539	-0.7774	-0.5841
Baseline	practical	largest four	100	-0.7202	0.02881	-0.828	-0.5899
Baseline	extended	full	50	-0.7243	0.008406	-0.7729	-0.6765
Baseline	extended	upper half	50	-0.7366	0.02608	-0.8169	-0.6584
Baseline	extended	largest four	50	-0.6349	0.03727	-0.7535	-0.522
Rich6D	practical	full	100	-0.6795	0.007969	-0.7368	-0.619
Rich6D	practical	upper half	100	-0.6474	0.02435	-0.7382	-0.5515
Rich6D	practical	largest four	100	-0.6826	0.02727	-0.7912	-0.5532
Rich6D	extended	full	50	-0.6952	0.007665	-0.74	-0.6491
Rich6D	extended	upper half	50	-0.7126	0.02493	-0.7894	-0.6357
Rich6D	extended	largest four	50	-0.6198	0.03905	-0.7347	-0.511

Table 7: Monte Carlo loss-oracle selection optimism, measured as cross-fitted minus plugin population regret. Pointwise percentile intervals resample whole paths and recalibrate both oracle rules in every draw; fold labels, anchors and population quadrature remain fixed. These intervals include calibration variability, unlike the separately retained descriptive paired-MCSE intervals.

Map	T	R	Crossfit minus plugin	Boot. low	Boot. high
Baseline	1440	100	0	0	0
Baseline	7290	50	0	0	1.917e-05
Rich6D	1440	100	0	0	0
Rich6D	7290	50	0	0	1.609e-05

Table 8: Practical-grid policy cells failing at least one predeclared numerical check. Every failing cell is retained here and in the machine-readable results. The extra horizons have no corresponding rank audit and remain outside the certified region regardless of quadrature status.

Map	Method	T	Floor / risk	Rank bound / risk	Rank pass	Quadrature pass
Baseline	holdout25	60	0.0001416	0.1292	False	True
Baseline	holdout25	360	0.000452	0.05323	False	True
Rich6D	holdout25	60	0.0007494	0.1187	False	True
Rich6D	rolling3	60	0.001534	0.05282	False	True
Rich6D	holdout25	360	0.002284	0.05838	False	True

Table 9: Baseline: sensitivity at the largest mean population loss on the fixed 96-penalty grid, in the 50-path numerical audit. Maximizing regret is equivalent because the analytic optimum is constant. The rank-1024 comparison uses the same penalty, without reselection. C512 and C1024 show proximity to the sample-rank boundary; payoff error is the squared difference on common quadrature. This is an ex-post diagnostic, not a tuning rule.

T	lambda	Max regret512	Same lambda regret1024	C512	C1024	Payoff error	Rank pass
60	1e-12	0.8037	0.6477	60	60	0.02426	False
90	1e-12	0.9623	0.7361	90	90	0.05084	False
120	1e-12	1.132	0.8287	120	120	0.08881	False
180	1e-12	1.431	0.9719	180	180	0.2023	False
240	1e-12	1.859	1.13	239.9	240	0.4105	False
360	1e-12	3.413	1.468	359.5	359.8	1.536	False
540	1e-12	12.07	2.173	504.8	539.2	9.843	False
720	1e-12	2.418	3.422	510.4	717.2	2.821	False
1080	1e-12	0.8888	8.708	511.1	994	7.831	False
1440	1e-12	0.5525	2.366	511.3	1016	1.805	False

Table 10: Rich6D: sensitivity at the largest mean population loss on the fixed 96-penalty grid, in the 50-path numerical audit. Maximizing regret is equivalent because the analytic optimum is constant. The rank-1024 comparison uses the same penalty, without reselection. C512 and C1024 show proximity to the sample-rank boundary; payoff error is the squared difference on common quadrature. This is an ex-post diagnostic, not a tuning rule.

T	lambda	Max regret512	Same lambda regret1024	C512	C1024	Payoff error	Rank pass
60	1e-12	0.8104	0.6546	60	60	0.02415	False
90	1e-12	0.9714	0.7424	90	90	0.05141	False
120	1e-12	1.122	0.8205	120	120	0.09049	False
180	1e-12	1.426	0.9637	180	180	0.2053	False
240	1e-12	1.867	1.129	239.9	240	0.4185	False
360	1e-12	3.46	1.456	359.5	359.8	1.574	False
540	1e-12	12.39	2.168	504.7	539.2	10.19	False
720	1e-12	2.419	3.384	510.4	717.2	2.784	False
1080	1e-12	0.8905	8.643	511.1	994	7.778	False
1440	1e-12	0.5546	2.412	511.3	1016	1.846	False

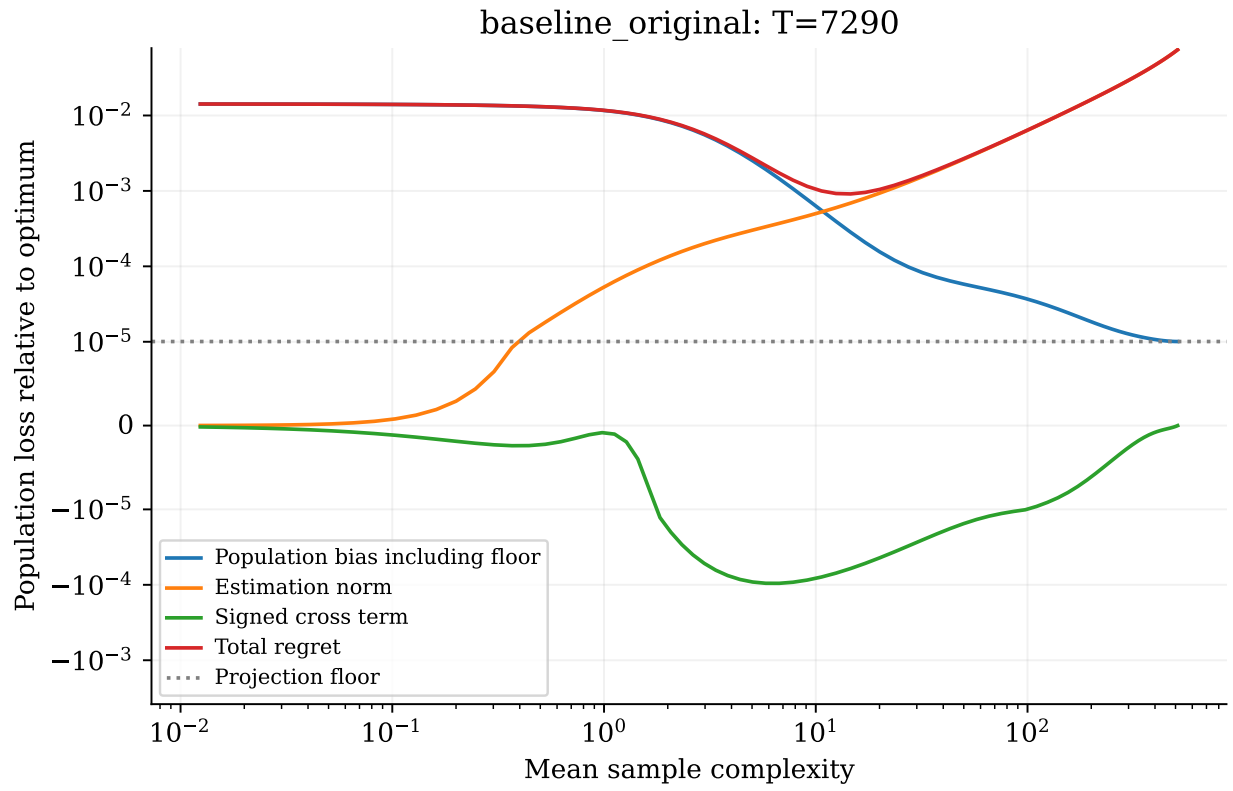


Figure 8: Baseline: regularization bias including projection floor, nonnegative estimation norm and signed cross term.

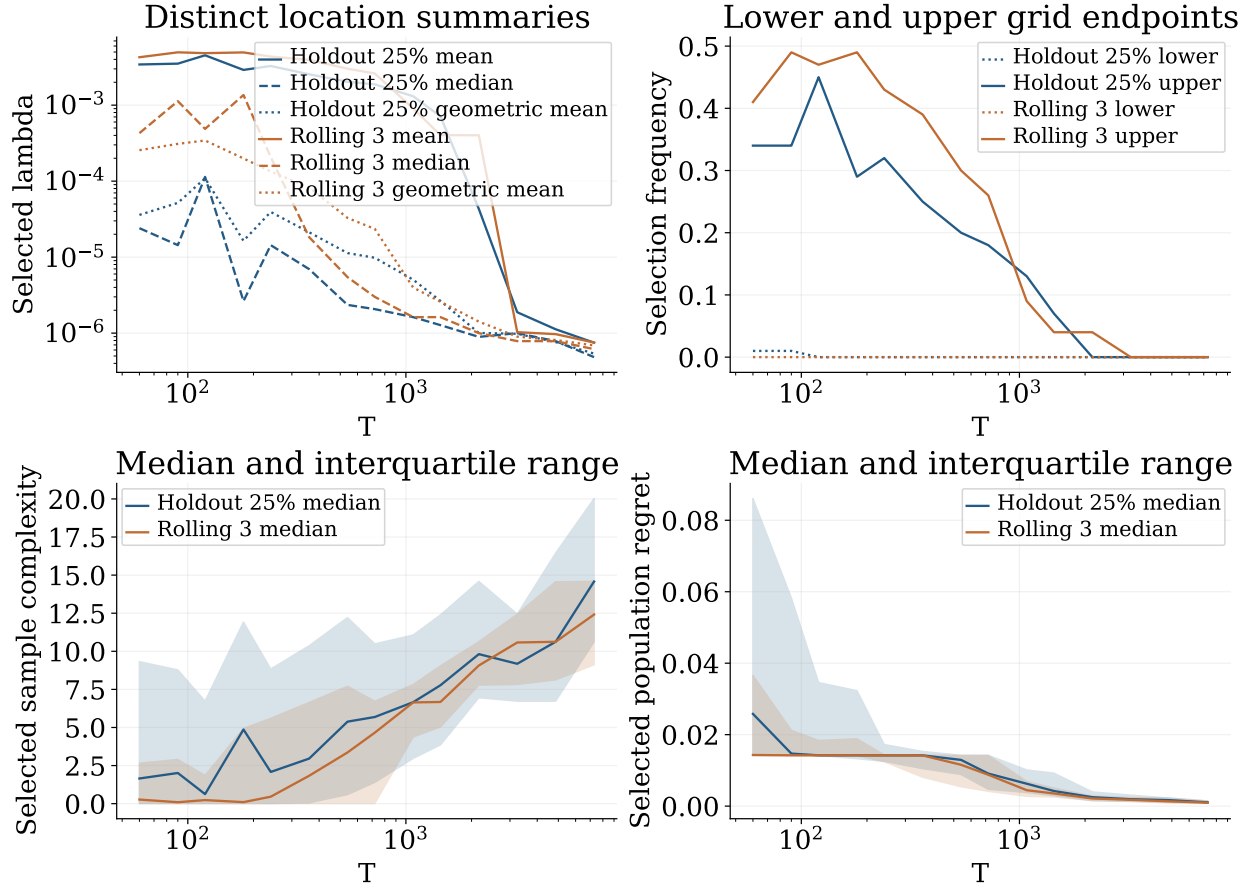


Figure 9: Baseline: separate penalty location summaries, endpoint frequencies, and median/interquartile ranges of complexity and regret.

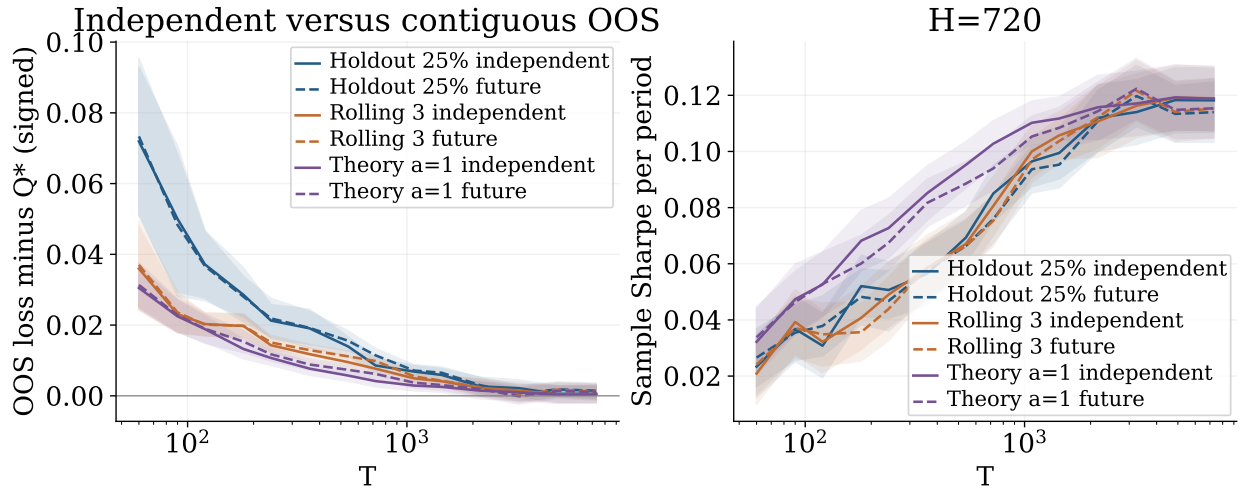


Figure 10: Baseline: independent stationary and contiguous future OOS. Negative sample excess losses remain visible.

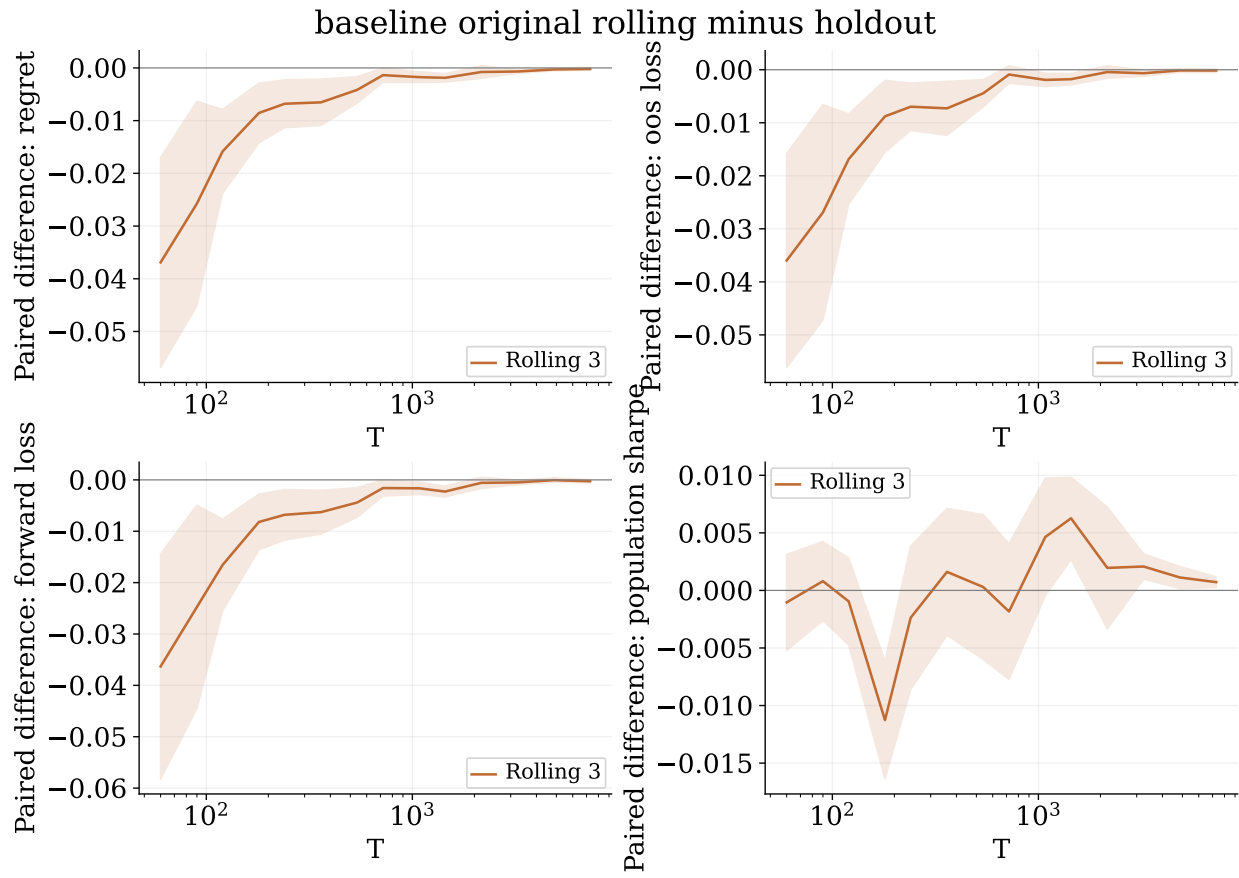


Figure 11: Baseline: paired rolling-minus-holdout differences; no selector was redesigned after observing confirmation outcomes.

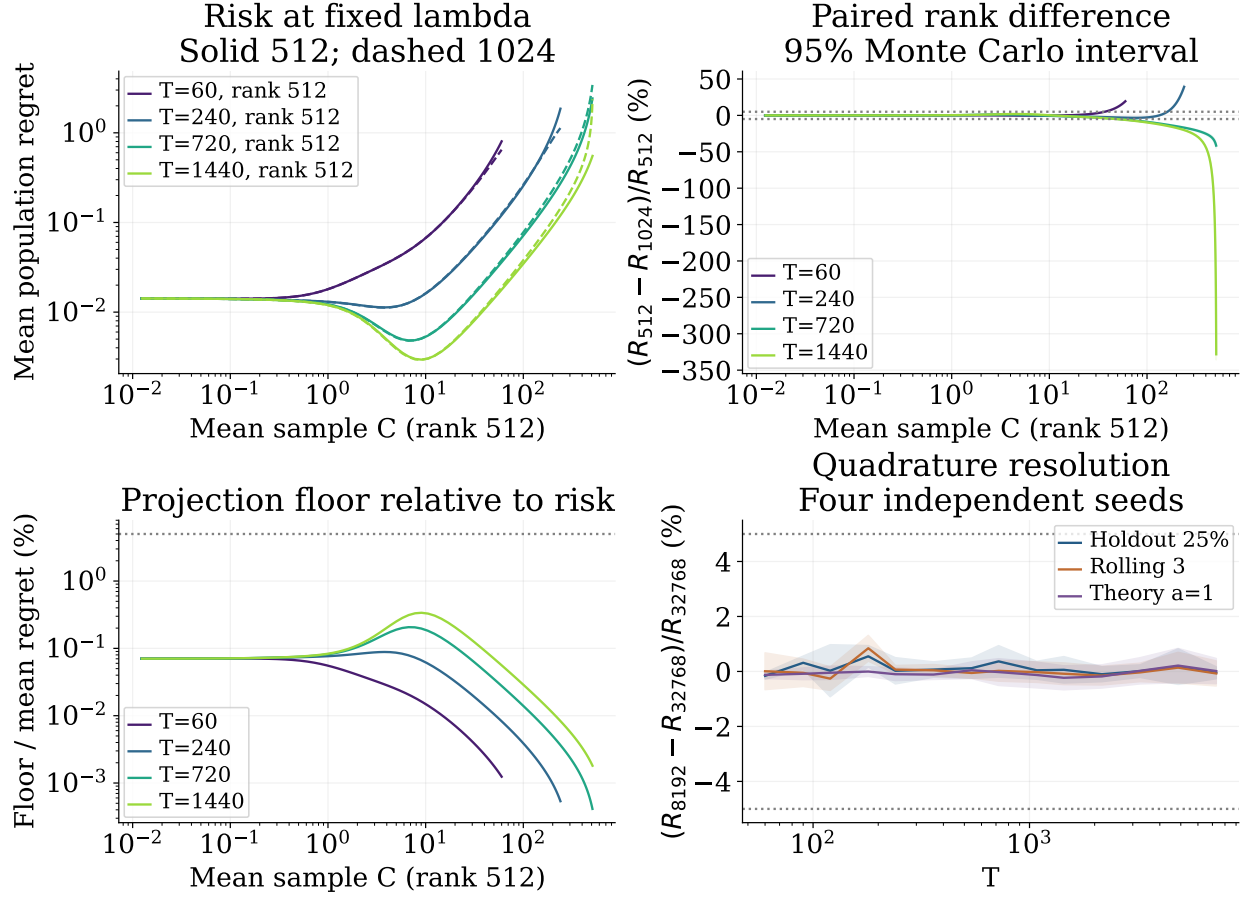


Figure 12: Baseline: fixed-penalty rank comparisons and independent quadrature sensitivity. Un-resolved cells are retained.

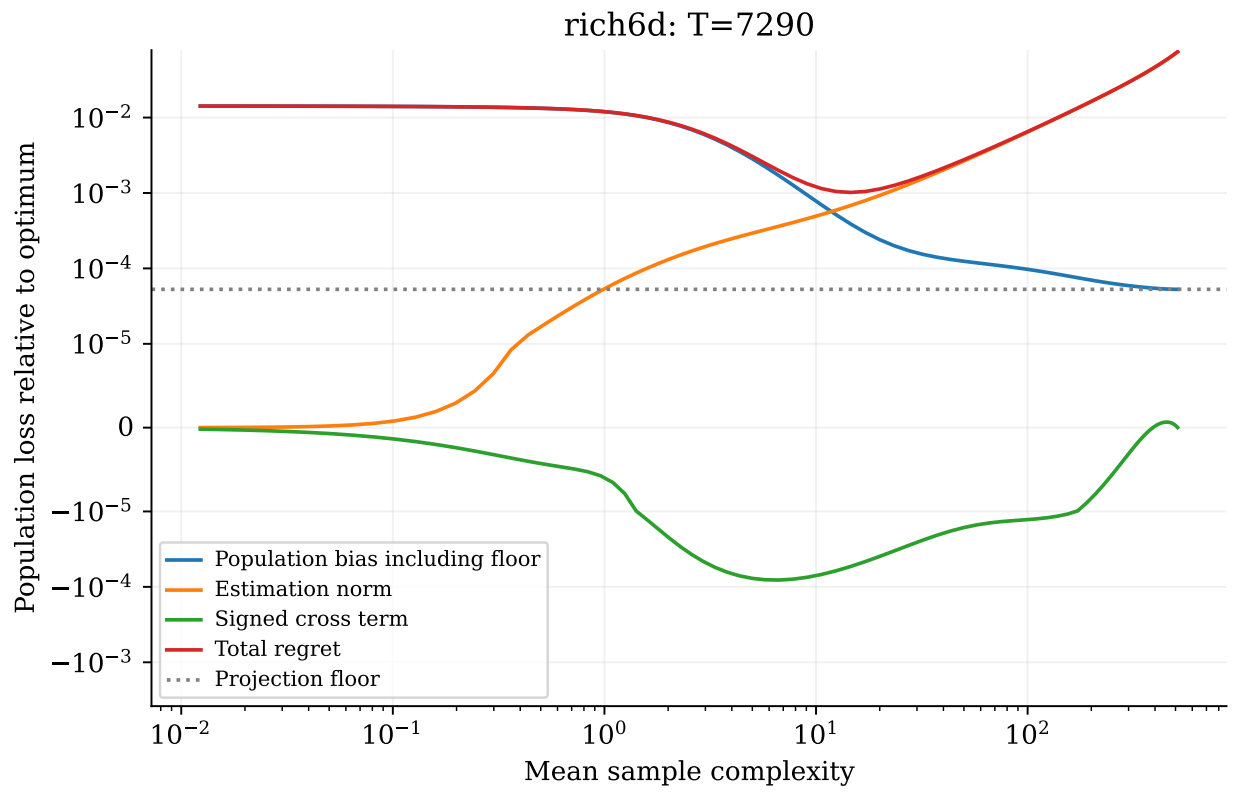


Figure 13: Rich6D: regularization bias including projection floor, nonnegative estimation norm and signed cross term.

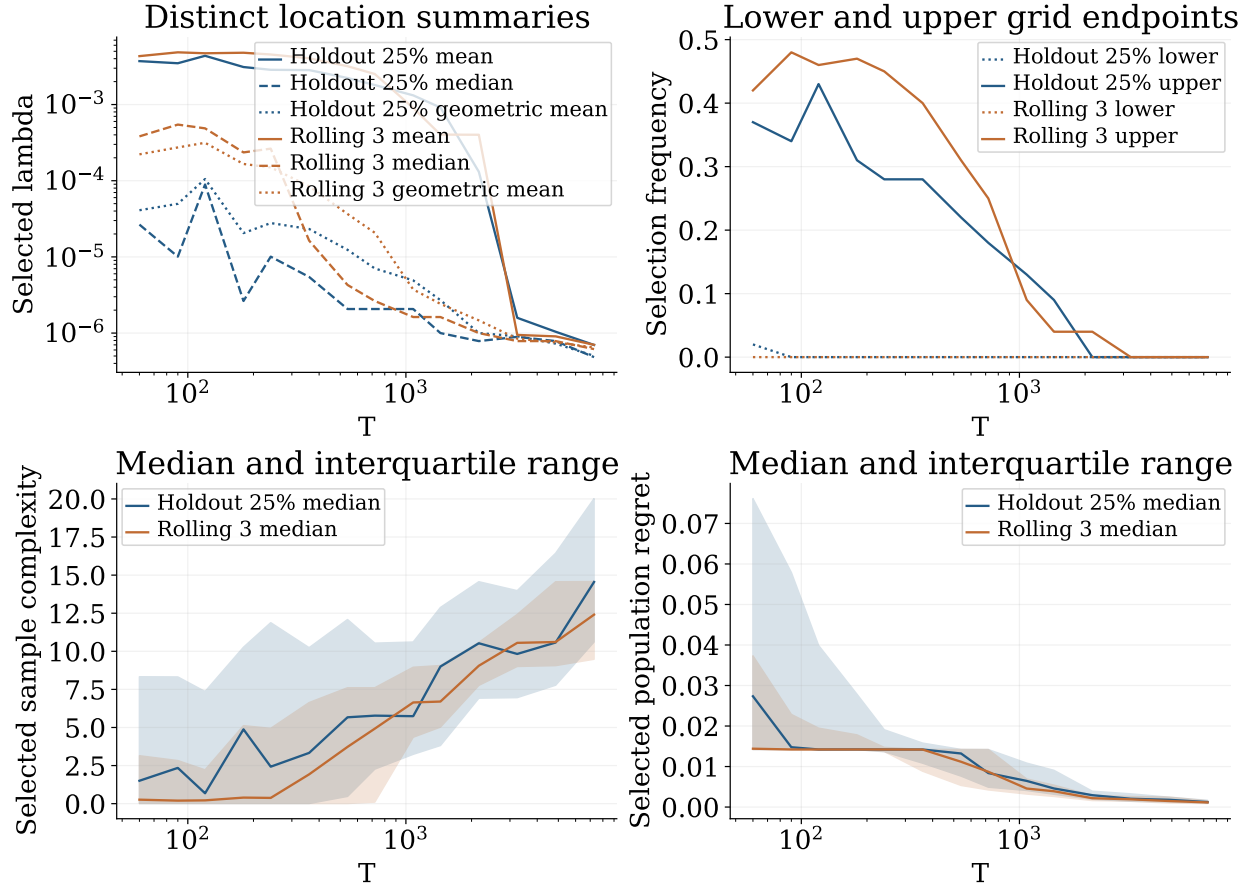


Figure 14: Rich6D: separate penalty location summaries, endpoint frequencies, and median/interquartile ranges of complexity and regret.

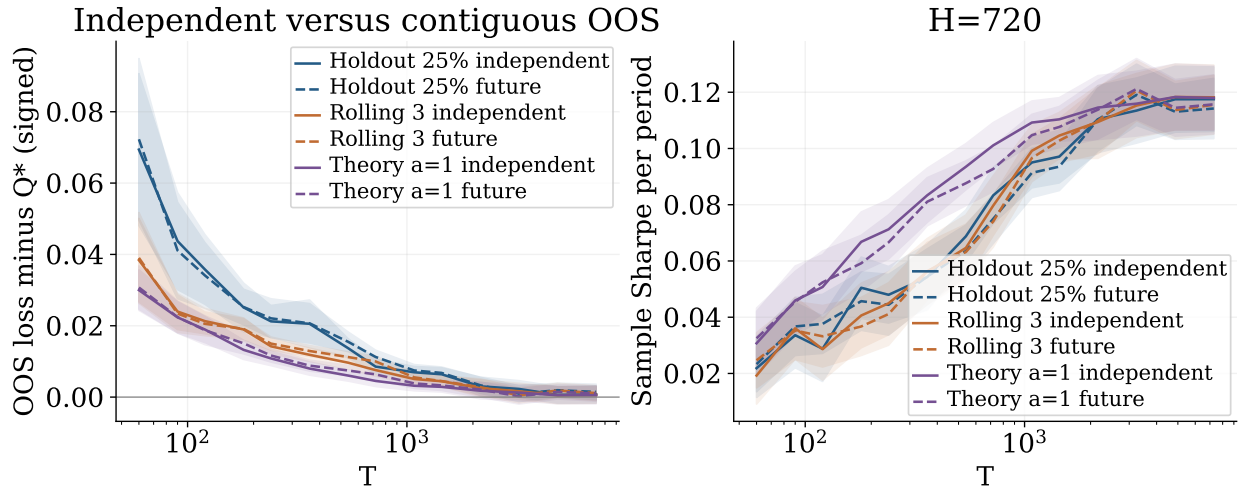


Figure 15: Rich6D: independent stationary and contiguous future OOS. Negative sample excess losses remain visible.

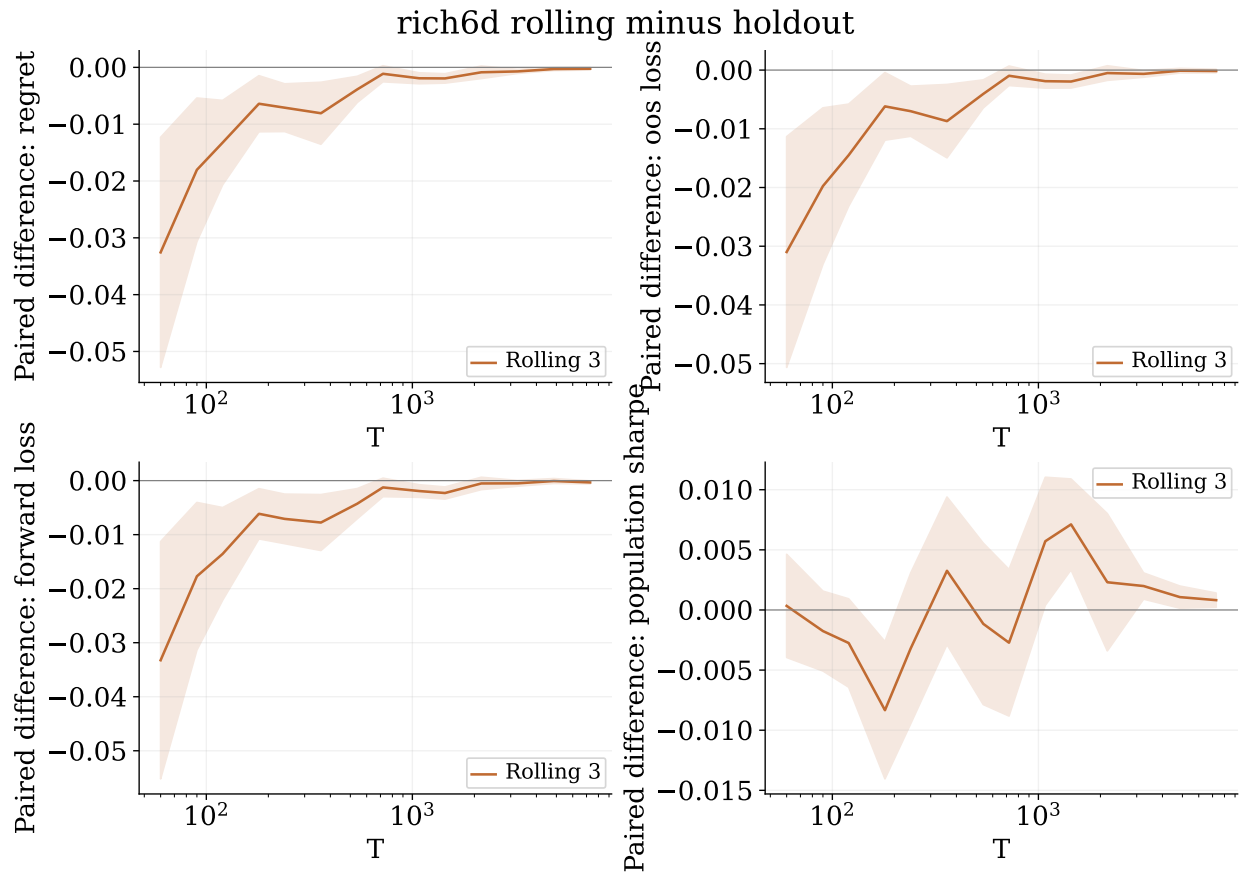


Figure 16: Rich6D: paired rolling-minus-holdout differences; no selector was redesigned after observing confirmation outcomes.

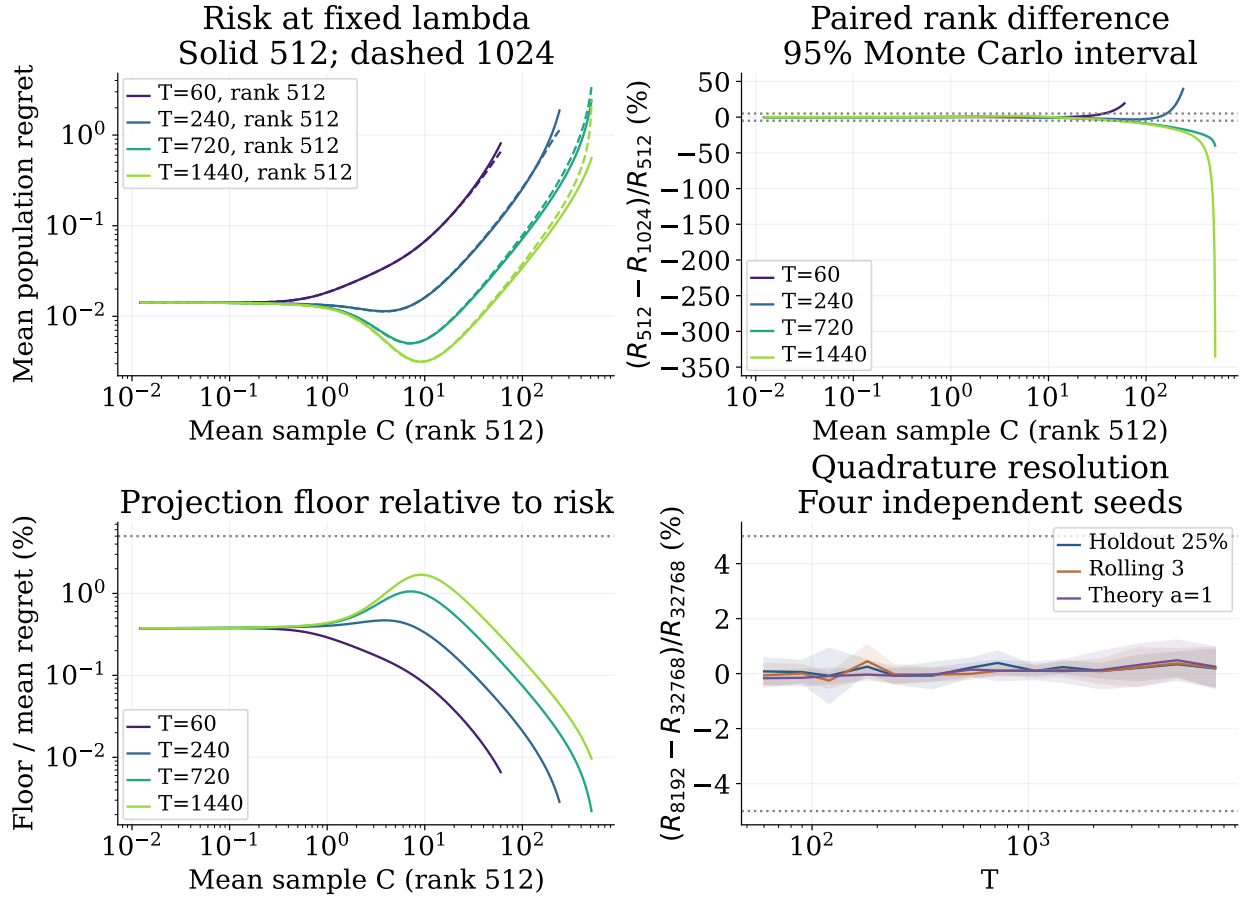


Figure 17: Rich6D: fixed-penalty rank comparisons and independent quadrature sensitivity. Unresolved cells are retained.

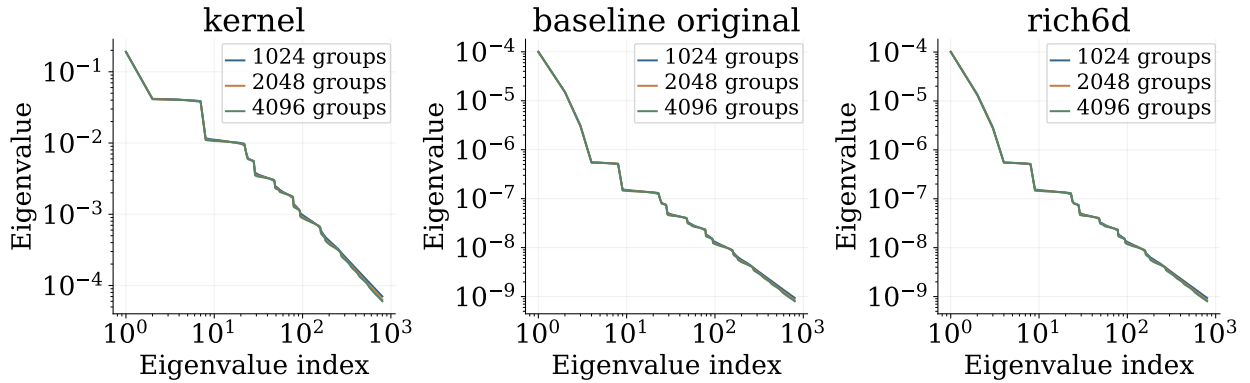


Figure 18: Three quadrature resolutions and three independent seeds. Shading spans the seed range, not a confidence interval. Fits use fixed windows and do not certify an asymptotic exponent.

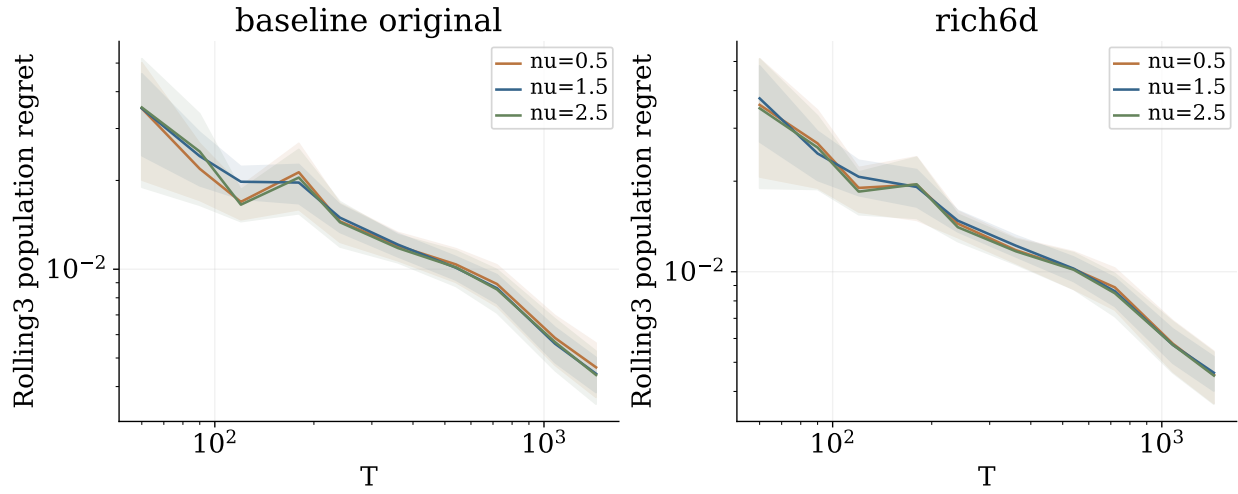


Figure 19: Prospective smoothness sensitivities on the practical grid; 50 paths per alternative map/kernel environment.

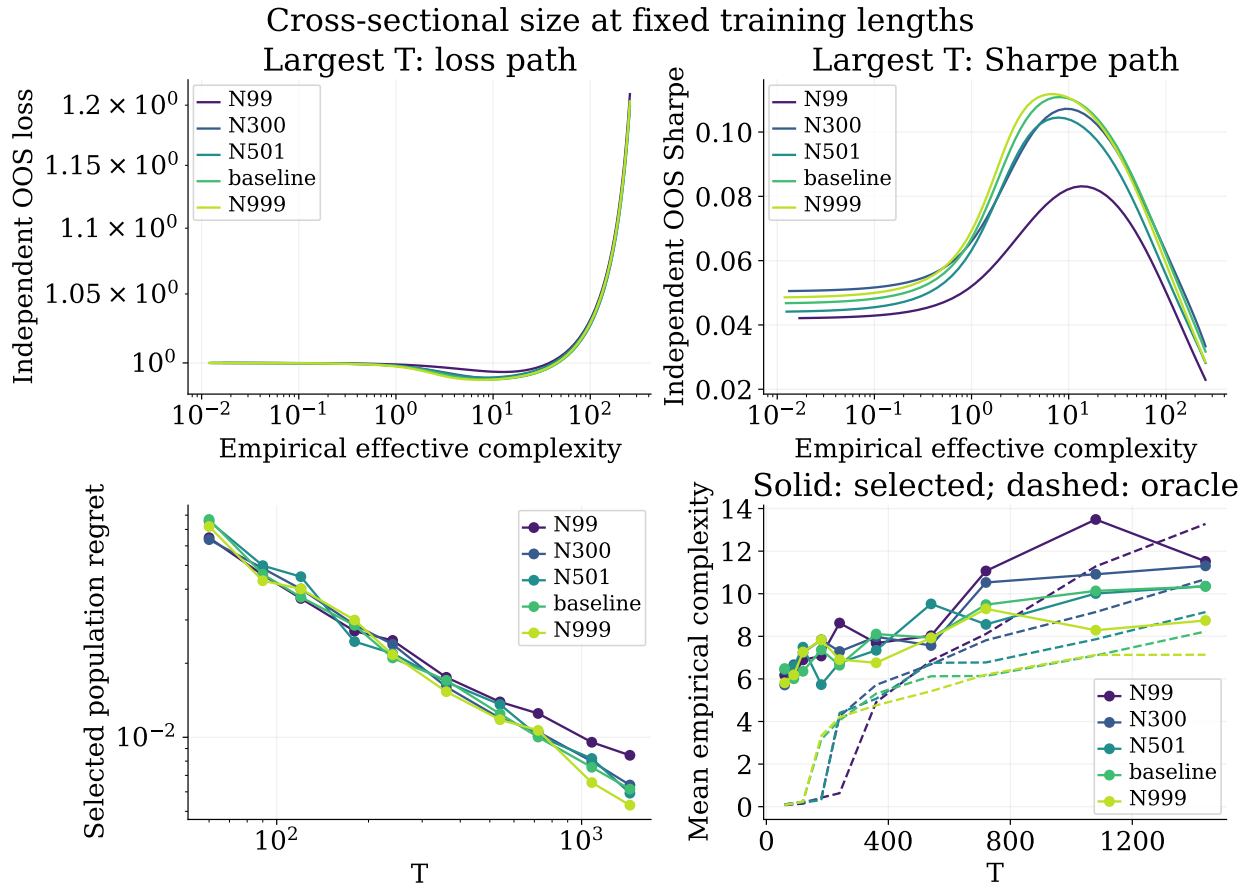


Figure 20: Historical cross-sectional-size sensitivity, archived unchanged and independently checked.

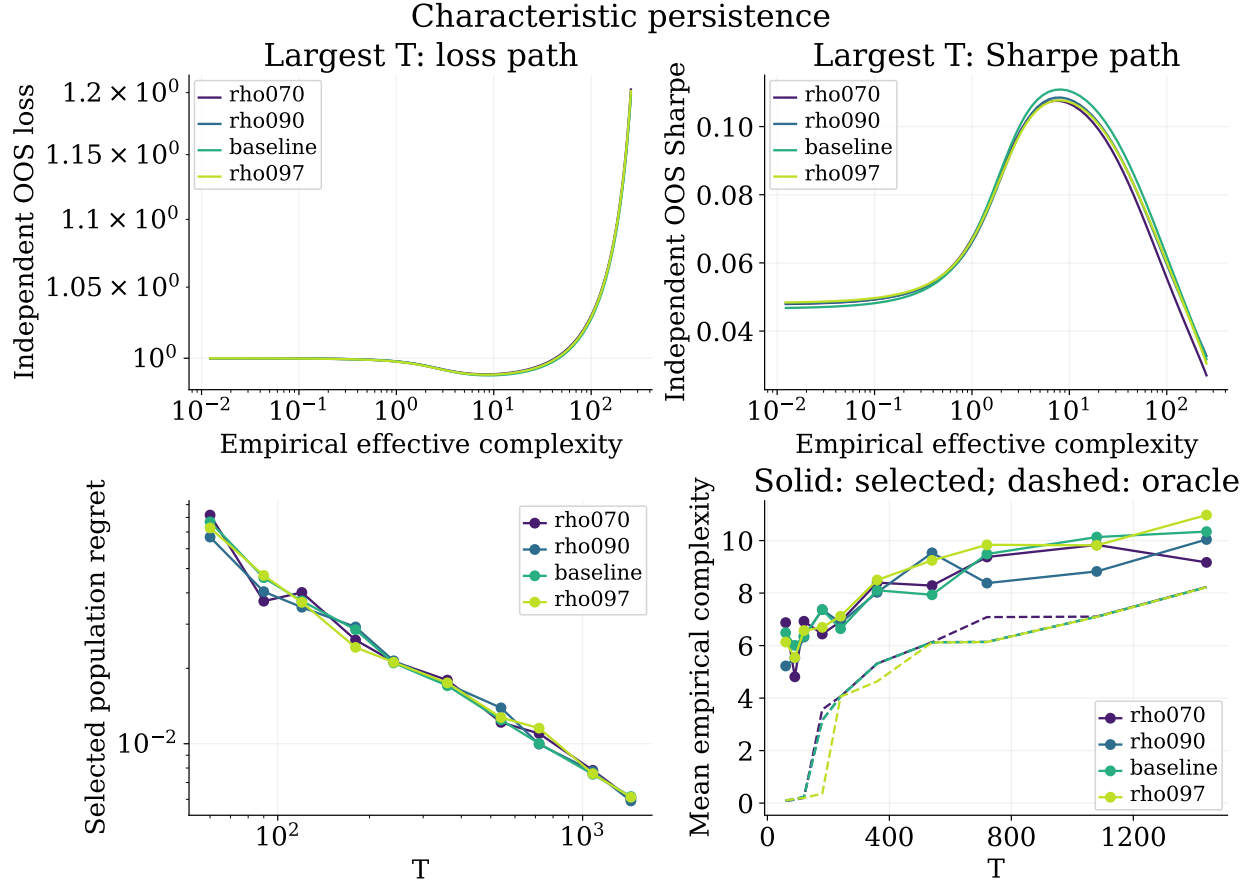


Figure 21: Historical characteristic-persistence sensitivity, archived unchanged and independently checked.